

Omada OCNA Practice Exam

Omada Certified Network Administrator - 84 multiple-choice questions

Instructions: Choose the single best answer for each question. Suggested time: 60 minutes. Aim for 70% or more. You can practise Part A and Part B separately depending on which OCNA track you take. The answer key with short explanations is on the last pages.

Note: These are original practice questions, not copies of real exam questions. They follow the publicly listed OCNA course outlines: the Wireless track (SDN introduction, device discovery/adoption/management, portal and hotspots, wireless fundamentals, demand analysis, network planning and optimization, troubleshooting) and the Routing and Switching track (network fundamentals, switching, routing, network services and security). According to training providers, the OCNA exam is taken online (about 60 minutes) and requires a score of 70% or higher. Port numbers, defaults and menu names can change between controller versions, so verify them against the official TP-Link Omada documentation.

PART A - Wireless Track (OCNA Wireless)

A1. Omada SDN Solution, Controller and Device Management

1. Which components form the Omada SDN solution?

- A) Cloud storage and EAP only
- B) Only EAP access points
- C) Omada Controller, EAP access points, switches and gateways
- D) VLAN-aware routers only

2. Which of these is NOT a type of Omada Controller?

- A) Hardware Controller (e.g. OC200)
- B) Software Controller
- C) Cloud-Based Controller
- D) Mobile-only Controller

3. Which TCP port is used by default for HTTPS management of the Omada Software Controller?

- A) 8088
- B) 8043
- C) 29810
- D) 443 only

4. Which UDP port does the Omada Controller use for device discovery?

- A) 67
- B) 8088
- C) 29810
- D) 29811

5. What is a Site in the Omada Controller?

- A) A VLAN
- B) A logical group of devices with its own settings, for example one branch or location
- C) A single wireless SSID
- D) A firmware version

6. What does adoption mean in Omada?

- A) A client is added to the guest list
- B) The firmware is downgraded
- C) The controller takes over management of a device and applies configuration to it
- D) A device is restored to factory defaults

7. What is a key characteristic of Standalone mode for an EAP?

- A) The AP is managed from a central cloud only
- B) The AP must use Mesh
- C) The AP cannot broadcast an SSID
- D) The AP is managed individually through its own web interface

8. What is the benefit of Omada Cloud-Based Controller?

- A) It replaces the gateway
- B) Centrally manage sites remotely without deploying local controller hardware
- C) It only works on a single LAN
- D) It removes the need for access points

9. Which hardware controller models belong to the Omada family?

- A) ER605 and ER7206
- B) TL-SG3428 and TL-SG2008
- C) OC200 and OC300
- D) EAP225 and EAP610

10. Which Omada guest portal type lets the administrator generate time- or data-limited codes?

- A) Simple Password
- B) MAC filtering
- C) No Authentication
- D) Voucher

11. What is the role of a captive portal?

- A) Perform firmware upgrades
- B) Redirect new clients to an authentication or splash page before granting internet access
- C) Encrypt wireless traffic with WPA3
- D) Balance WAN links

12. Which technology provides port-based network access control using a RADIUS server?

- A) 802.11r
- B) 802.1Q
- C) 802.3ad
- D) 802.1X

A2. Portal and Hotspot

13. Which setting lets unauthenticated clients reach specific destinations, such as a payment or login page, before passing the portal?

- A) Fast roaming
- B) Port mirroring
- C) Pre-authentication access
- D) Band steering

14. What typically happens when a guest's voucher time limit expires?

- A) The client keeps access until it reboots
- B) The AP reboots
- C) The client is logged out and must authenticate again with a valid voucher
- D) The SSID is deleted

15. Why is a guest SSID usually mapped to a separate VLAN?

- A) To increase the 2.4 GHz range
- B) To enable PoE for clients
- C) To disable DHCP for guests
- D) To isolate guests from internal resources and apply separate policies

A3. Wireless Fundamentals

16. Which three channels are non-overlapping in the 2.4 GHz band with 20 MHz width?

- A) 1, 6 and 11
- B) 1, 5 and 9
- C) 2, 7 and 12
- D) 3, 6 and 9

17. Which IEEE standard is marketed as Wi-Fi 6?

- A) 802.11ax
- B) 802.11ad
- C) 802.11ac
- D) 802.11n

18. Which frequency band was added by Wi-Fi 6E?

- A) 60 GHz
- B) 6 GHz
- C) 900 MHz
- D) 3.6 GHz

19. Which technology lets an AP serve multiple clients within a single transmission by dividing the channel into resource units?

- A) Band steering
- B) OFDMA
- C) STP
- D) CSMA/CD

20. Which Wi-Fi 5 (802.11ac) band is used?

- A) Both 2.4 GHz and 5 GHz exclusively
- B) 2.4 GHz only
- C) 6 GHz only
- D) 5 GHz only

21. Which statement about 2.4 GHz compared to 5 GHz is correct?

- A) 2.4 GHz has better range and wall penetration but is more congested
- B) 2.4 GHz offers higher maximum throughput
- C) 2.4 GHz has more non-overlapping channels
- D) 2.4 GHz is not affected by microwave ovens

22. What does WPA3-Personal use to replace the PSK four-way handshake weakness against offline dictionary attacks?

- A) WEP
- B) SAE (Simultaneous Authentication of Equals)
- C) MAC filtering
- D) TKIP

23. What is the purpose of band steering?

- A) Encrypt traffic between APs
- B) Move clients to the 2.4 GHz band
- C) Change the SSID automatically
- D) Encourage dual-band clients to connect to the 5 GHz band

24. Which technology allows an AP to transmit to multiple clients simultaneously using spatial streams?

- A) Beamforming only
- B) MU-MIMO
- C) Channel bonding
- D) SU-MIMO

25. Which 802.11 amendment defines Fast BSS Transition (fast roaming)?

- A) 802.11s
- B) 802.11r
- C) 802.11e
- D) 802.11w

26. What is the effect of using a wider channel width (e.g. 80 MHz instead of 20 MHz)?

- A) No effect on throughput
- B) Longer range and fewer interference problems
- C) Lower throughput but better security
- D) Higher potential throughput but fewer non-overlapping channels and more interference

27. In a wireless mesh deployment, what is a wireless uplink?

- A) An AP connecting to another AP over the air instead of Ethernet
- B) A connection between an AP and a wired switch
- C) A PoE injector connected to an AP
- D) A client connected to an AP via WPS

A4. Demand Analysis, Network Planning and Optimization

28. An office expects 100 wireless clients, about 50% active at the same time, each needing about 2 Mbps. What aggregate throughput should be planned for?

- A) 100 Mbps
- B) 200 Mbps
- C) 50 Mbps
- D) 1 Gbps

29. Which information is most important to gather during wireless demand analysis?

- A) Only the colour of the building
- B) Only the price of the access points
- C) Only the ISP's marketing name
- D) Number of clients, application/bandwidth needs, coverage area and environment

30. What is the main goal of a wireless site survey?

- A) Replace the DHCP server
- B) Determine AP placement, coverage and sources of interference
- C) Configure VLAN tagging on switches
- D) Create guest vouchers

31. What problem occurs when neighboring APs use the same channel?

- A) PoE overload
- B) Spanning tree loops
- C) DNS poisoning
- D) Co-channel interference (clients and APs compete for the same airtime)

32. What is a typical side effect of setting every AP to maximum transmit power in a dense deployment?

- A) More co-channel interference and clients sticking to distant APs
- B) Lower power consumption
- C) Faster roaming for all clients
- D) Automatic band steering

33. Which 5 GHz channels may have to be vacated when radar is detected?

- A) Channels reserved for Bluetooth
- B) Channels 1, 6 and 11
- C) DFS channels
- D) All 2.4 GHz channels

34. How should AP coverage cells be designed to support roaming?

- A) Cells should not overlap at all
- B) Each AP should use a different SSID
- C) Adjacent cells should overlap slightly
- D) Cells should fully overlap on the same channel

A5. Troubleshooting

- 35. A client connects to Wi-Fi but gets an address like 169.254.x.x. What is the most likely cause?**
- A) The DNS server is down
 - B) The AP uses WPA3
 - C) The client received no response from a DHCP server
 - D) The client uses 5 GHz
- 36. A client can ping 8.8.8.8 but cannot open websites by name. What is the most likely problem?**
- A) DHCP lease
 - B) PoE budget
 - C) DNS resolution
 - D) VLAN trunking
- 37. Which Windows command shows the IP address, default gateway and DNS servers of an adapter in detail?**
- A) ipconfig /all
 - B) netstat -r only
 - C) tracert
 - D) nslookup -ip
- 38. Which Windows command shows the path packets take to a destination hop by hop?**
- A) ipconfig /renew
 - B) arp -d
 - C) route add
 - D) tracert
- 39. An adopted AP shows as Disconnected in the controller. What is the best first check?**
- A) Power/PoE, cable link and IP connectivity to the controller
 - B) Change the SSID name
 - C) Rebuild the guest portal
 - D) Switch the AP to the 6 GHz band
- 40. In bottom-up troubleshooting, which OSI layer is checked first?**
- A) Physical layer
 - B) Network layer
 - C) Session layer
 - D) Application layer

PART B - Routing and Switching Track (OCNA R&S;)

B1. Network Fundamentals

- 41. Which OSI layer is responsible for routing packets between different networks?**
- A) Network layer (Layer 3)
 - B) Transport layer (Layer 4)
 - C) Data Link layer (Layer 2)
 - D) Session layer (Layer 5)
- 42. A standard Layer 2 switch makes forwarding decisions based on which address?**
- A) Hostname
 - B) Source port number
 - C) Destination MAC address
 - D) Destination IP address
- 43. How many usable host addresses are available in a /26 subnet?**
- A) 126
 - B) 30
 - C) 62
 - D) 64

44. Which of the following is a private IPv4 address?

- A) 192.169.1.1
- B) 11.0.0.1
- C) 172.20.5.10
- D) 172.32.0.1

45. What is the correct order of the DHCP process?

- A) Discover, Offer, Request, Acknowledge
- B) Offer, Discover, Acknowledge, Request
- C) Discover, Request, Offer, Acknowledge
- D) Request, Discover, Acknowledge, Offer

46. Which protocol resolves an IP address to a MAC address on a local network?

- A) ICMP
- B) DNS
- C) DHCP
- D) ARP

47. Which transport protocol is connectionless and does not guarantee delivery?

- A) UDP
- B) TCP
- C) TLS
- D) SCTP

48. What is the primary purpose of NAT?

- A) Encrypt traffic between sites
- B) Translate private IP addresses to public IP addresses
- C) Resolve domain names
- D) Assign IP addresses automatically

49. What is the network address of host 192.168.10.77/26?

- A) 192.168.10.64
- B) 192.168.10.0
- C) 192.168.10.76
- D) 192.168.10.128

50. Which protocol does the ping command use?

- A) ICMP
- B) SNMP
- C) ARP
- D) UDP port 53

51. Which well-known port is used by DNS?

- A) 443
- B) 80
- C) 67
- D) 53

52. What does the default gateway do for a host?

- A) Assigns IP addresses to clients
- B) Translates domain names
- C) Blocks all inbound traffic
- D) Forwards traffic destined for other networks

B2. Switching Basics and Technologies

53. Which IEEE standard defines VLAN tagging?

- A) IEEE 802.3ad
- B) IEEE 802.1Q
- C) IEEE 802.11ac
- D) IEEE 802.1X

54. What is the valid range of usable VLAN IDs?

- A) 1 to 1024
- B) 1 to 255
- C) 0 to 4095
- D) 1 to 4094

55. Which port type carries traffic for multiple VLANs between switches?

- A) Access port
- B) Mirror port
- C) Isolated port
- D) Trunk port

56. What is the main purpose of Spanning Tree Protocol (STP)?

- A) Provide PoE to devices
- B) Encrypt Layer 2 frames
- C) Increase port bandwidth
- D) Prevent Layer 2 loops

57. Which standard defines Rapid Spanning Tree Protocol (RSTP)?

- A) IEEE 802.1Q
- B) IEEE 802.1D
- C) IEEE 802.3af
- D) IEEE 802.1w

58. Which feature combines multiple physical links into one logical link for more bandwidth and redundancy?

- A) Link Aggregation (LACP, 802.3ad)
- B) IGMP snooping
- C) VLAN tagging
- D) Port mirroring

59. Which PoE standard delivers up to 30 W per port at the power sourcing equipment?

- A) IEEE 802.3at (PoE+)
- B) IEEE 802.3af (PoE)
- C) IEEE 802.1p
- D) IEEE 802.3u

60. What does port mirroring do?

- A) Duplicates the MAC table to a second switch
- B) Doubles the bandwidth of a port
- C) Copies traffic from one port to another for analysis
- D) Blocks traffic on unused ports

61. What is the benefit of IGMP snooping?

- A) Limits multicast traffic to ports that requested it
- B) Assigns IPs to multicast groups
- C) Prevents VLAN hopping
- D) Detects cable faults

62. Which feature limits excessive broadcast, multicast or unknown unicast traffic on a port?

- A) Link backup
- B) Storm Control
- C) Port isolation
- D) DHCP snooping

63. What is the native VLAN on a trunk port?

- A) The VLAN that is always blocked
- B) The VLAN with the highest ID
- C) The VLAN whose traffic is sent untagged
- D) The VLAN used only for management

64. Which feature detects when a cable creates a loop from a switch port back into the same switch?

- A) Rate limiting
- B) Loopback detection
- C) Port mirroring
- D) QoS

B3. Routing Basics and Features

65. A router has routes 10.1.0.0/16 and 10.1.1.0/24. Which route is used for destination 10.1.1.5?

- A) The default route
- B) 10.1.0.0/16
- C) Both are load-balanced
- D) 10.1.1.0/24

66. What does the default route 0.0.0.0/0 represent?

- A) Only multicast traffic
- B) Any destination not matched by a more specific route
- C) Only the loopback address
- D) Only the local subnet

67. What is required for communication between two different VLANs?

- A) A Layer 1 hub
- B) A Layer 3 device (router or Layer 3 switch)
- C) Port mirroring
- D) A wireless repeater

68. Which statement about static routes is correct?

- A) They are configured manually and do not adapt automatically to topology changes
- B) They are learned automatically from neighbors
- C) They only work with IPv6
- D) They require OSPF

69. OSPF is which type of routing protocol?

- A) Layer 2 loop prevention protocol
- B) Link-state interior gateway protocol
- C) Authentication protocol
- D) Distance-vector exterior gateway protocol

70. What must be true for the next hop of a static route?

- A) It must be a public IP address
- B) It must be reachable on a directly connected network
- C) It must be a DNS server
- D) It must be in a different VLAN than the router

B4. Network Services and Security

71. Which feature lets an Omada Gateway use several WAN connections at the same time?

- A) STP
- B) Band steering
- C) WAN load balancing
- D) Port mirroring

72. In Link Backup mode, when is the backup WAN used?

- A) Only for guest traffic
- B) Only at night
- C) When the primary WAN link fails
- D) Always, at the same time as the primary

73. What does policy routing allow?

- A) Blocking all unknown MAC addresses
- B) Sending specific traffic over a chosen WAN or next hop based on defined rules
- C) Creating PoE schedules
- D) Assigning VLAN IDs automatically

74. Which VPN type connects two branch offices permanently?

- A) Captive portal VPN
- B) Guest VPN
- C) Client-to-Site VPN
- D) Site-to-Site VPN

75. Which protocol suite is commonly used to build encrypted site-to-site VPN tunnels?

- A) ICMP
- B) ARP
- C) IPsec
- D) DHCP

76. What is the purpose of an ACL on an Omada Gateway?

- A) Permit or deny traffic between networks based on defined rules
- B) Increase wireless range
- C) Provide PoE
- D) Update device firmware

77. What is a DMZ typically used for?

- A) Increasing WiFi speed
- B) Assigning guest vouchers
- C) Hosting public-facing servers separated from the internal LAN
- D) Bundling ports

78. What does port forwarding (virtual server) do?

- A) Bundles two WAN ports
- B) Copies traffic to a monitoring port
- C) Disables NAT
- D) Maps an external port on the WAN to an internal host and port

79. What is the advantage of using VLANs for segmentation, such as separate Guest and IoT networks?

- A) Removes the need for a gateway
- B) Isolates broadcast domains and allows security policies per network
- C) Automatically encrypts all traffic
- D) Increases the number of WAN ports

80. Which protocol is commonly used to collect device monitoring information such as interface counters or CPU load?

- A) SMTP
- B) SNMP
- C) NTP
- D) TFTP

81. Which protocol keeps device clocks synchronized so that logs have consistent timestamps?

- A) NTP
- B) SNMP
- C) DHCP
- D) SSH

82. What is the benefit of DHCP reservation (address binding)?

- A) It disables DHCP on the network
- B) It increases lease time to infinite for all clients
- C) It encrypts DHCP messages
- D) A client always receives the same IP address based on its MAC address

83. Which feature helps prevent rogue DHCP servers on a switch network?

- A) Port mirroring
- B) DHCP snooping
- C) Link aggregation
- D) IGMP snooping

84. Which feature allows DHCP clients in one network to get addresses from a DHCP server in another network?

- A) NAT hairpinning
- B) DHCP reservation
- C) DHCP relay
- D) Link backup

Answer Key

#	Ans	#	Ans	#	Ans	#	Ans
1	C	22	B	43	C	64	B
2	D	23	D	44	C	65	D
3	B	24	B	45	A	66	B
4	C	25	B	46	D	67	B
5	B	26	D	47	A	68	A
6	C	27	A	48	B	69	B
7	D	28	A	49	A	70	B
8	B	29	D	50	A	71	C
9	C	30	B	51	D	72	C
10	D	31	D	52	D	73	B
11	B	32	A	53	B	74	D
12	D	33	C	54	D	75	C
13	C	34	C	55	D	76	A
14	C	35	C	56	D	77	C
15	D	36	C	57	D	78	D
16	A	37	A	58	A	79	B
17	A	38	D	59	A	80	B
18	B	39	A	60	C	81	A
19	B	40	A	61	A	82	D
20	D	41	A	62	B	83	B
21	A	42	C	63	C	84	C

Explanations

1. **C** - Omada integrates network devices under a central management platform.
2. **D** - The types are Hardware, Software and Cloud-Based controllers.
3. **B** - 8088 is HTTP management, 8043 is HTTPS management, 8843 is the portal HTTPS port.
4. **C** - 29810/UDP discovery; 29811-29814/TCP are used for manage, adopt, upgrade and manage-v2.
5. **B** - Sites allow managing multiple locations from one controller.
6. **C** - Devices appear as Pending until adopted.
7. **D** - Standalone is suitable for small deployments with a few devices.
8. **B** - Devices connect to the cloud, so remote management over the internet is possible.
9. **C** - ER = gateways, TL-SG = switches, EAP = access points.
10. **D** - Vouchers can be printed or distributed to guests with limits on duration and traffic.
11. **B** - Typically used in hotels, cafes and guest networks.
12. **D** - 802.1X authenticates devices or users before access to the LAN or WLAN is granted.
13. **C** - Pre-authentication access whitelists resources that must be reachable before authentication.
14. **C** - Vouchers limit duration and/or data; after expiry the portal is shown again.
15. **D** - Segmentation keeps guest traffic away from internal servers and printers.
16. **A** - Channels are spaced 5 MHz apart but 20 MHz wide, so 1/6/11 do not overlap.
17. **A** - 802.11ax = Wi-Fi 6 (and Wi-Fi 6E in the 6 GHz band). 802.11ac = Wi-Fi 5, 802.11n = Wi-Fi 4.
18. **B** - Wi-Fi 6E extends Wi-Fi 6 into the 6 GHz band.
19. **B** - OFDMA is a key Wi-Fi 6 feature that improves efficiency, especially for many small packets.
20. **D** - 802.11ac operates only on 5 GHz; dual-band Wi-Fi 5 APs use 802.11n on 2.4 GHz.
21. **A** - Lower frequencies travel farther but offer fewer channels and more interference.
22. **B** - SAE provides forward secrecy and protects against offline brute force.
23. **D** - 5 GHz is typically less congested and faster.
24. **B** - Multi-User MIMO serves several clients at the same time.
25. **B** - 802.11r reduces handoff time; 802.11k/v assist with neighbor reports and steering.
26. **D** - Wider channels trade capacity per client against channel reuse.
27. **A** - Mesh APs relay traffic wirelessly, reducing cabling but also available bandwidth.
28. **A** - 100 clients x 50% concurrency x 2 Mbps = 100 Mbps.
29. **D** - Capacity (clients and applications) and coverage (area and obstacles) drive AP count and placement.
30. **B** - Surveys can be predictive (planning tools) or on-site (measurements).
31. **D** - Use channel planning, e.g. 1/6/11 on 2.4 GHz, to reduce co-channel interference.
32. **A** - Lower, balanced power creates smaller cells and better capacity and roaming.
33. **C** - Dynamic Frequency Selection protects radar systems; non-DFS channels are more stable.
34. **C** - A small overlap allows clients to hand off without losing connection.
35. **C** - 169.254.0.0/16 is the APIPA (link-local) range.
36. **C** - IP connectivity works, so name resolution is failing.
37. **A** - ipconfig /all lists complete adapter configuration including DHCP and DNS.
38. **D** - Traceroute (tracert on Windows) uses TTL/ICMP to list each hop.
39. **A** - Start with the physical layer and basic reachability before touching configuration.
40. **A** - Cables, power, link LEDs and ports come first.
41. **A** - Routers forward packets using IP addresses at Layer 3.
42. **C** - Switches build a MAC address table and forward frames by destination MAC.
43. **C** - /26 leaves 6 host bits: $2^6 - 2 = 62$.

- 44. **C** - Private ranges: 10.0.0.0/8, 172.16.0.0/12 (172.16-172.31), 192.168.0.0/16.
- 45. **A** - Remembered as DORA.
- 46. **D** - ARP maps IPv4 addresses to MAC addresses within a broadcast domain.
- 47. **A** - UDP has no handshake or retransmission; it is used for DNS queries, VoIP, streaming.
- 48. **B** - NAT lets many private hosts share one or few public addresses.
- 49. **A** - /26 blocks are 64 addresses: 0, 64, 128, 192. 77 falls into the 64-127 block.
- 50. **A** - Ping sends ICMP Echo Request and receives Echo Reply.
- 51. **D** - DNS uses port 53 (UDP and TCP). 67 is DHCP server, 80 HTTP, 443 HTTPS.
- 52. **D** - Traffic outside the local subnet is sent to the default gateway.
- 53. **B** - 802.1Q inserts a 4-byte VLAN tag into the Ethernet frame.
- 54. **D** - IDs 0 and 4095 are reserved.
- 55. **D** - Trunk ports carry tagged frames of several VLANs; access ports belong to one VLAN.
- 56. **D** - STP blocks redundant paths to avoid broadcast storms.
- 57. **D** - 802.1w converges much faster than the original 802.1D STP.
- 58. **A** - LAG/LACP bundles ports; load is balanced per flow.
- 59. **A** - 802.3af = 15.4 W, 802.3at = 30 W, 802.3bt = 60/90 W.
- 60. **C** - Used with a sniffer or IDS connected to the destination port.
- 61. **A** - Without it, multicast is flooded to all ports in the VLAN.
- 62. **B** - Storm Control sets thresholds to protect the network from floods.
- 63. **C** - Untagged frames received on a trunk are assigned to the native VLAN (PVID).
- 64. **B** - The port is blocked or shut down to prevent a broadcast storm.
- 65. **D** - The longest prefix match wins.
- 66. **B** - It is typically pointed to the ISP or upstream gateway.
- 67. **B** - Inter-VLAN routing can be done with a router (router-on-a-stick) or an L3 switch.
- 68. **A** - Dynamic protocols such as OSPF adapt; static routes are simple and predictable.
- 69. **B** - OSPF routers exchange link-state information within an autonomous system.
- 70. **B** - The router must be able to resolve the next hop on a connected interface.
- 71. **C** - Load balancing distributes traffic; link backup is used for failover.
- 72. **C** - Link Backup provides failover, not simultaneous load sharing.
- 73. **B** - Rules match source/destination IPs or services and select an outbound interface.
- 74. **D** - Site-to-Site tunnels link entire networks; Client-to-Site is for individual remote users.
- 75. **C** - IPsec provides authentication and encryption at the network layer.
- 76. **A** - Gateway ACLs filter traffic for example between VLANs or to the WAN.
- 77. **C** - A DMZ limits exposure of the internal network if a public server is compromised.
- 78. **D** - Used to publish internal services such as a web server.
- 79. **B** - Segmentation reduces attack surface and broadcast traffic.
- 80. **B** - SNMP is used by network management systems for monitoring.
- 81. **A** - NTP synchronizes time with a server.
- 82. **D** - Useful for printers, servers and cameras.
- 83. **B** - Only ports marked trusted may send DHCP server messages.
- 84. **C** - A relay agent forwards DHCP broadcasts as unicast to the server.